Cart-page upsell nudges

Adding a 'Frequently bought with items in your cart' module on the cart page increases average order value through additional complementary purchases.

AT A GLANCE

Primary metric is *conversion_rate*; decision is binary against Ship if AOV CI excludes zero on benefit side.

Pre-registered before any analysis. Decision is binary against conversion_rate: a 2.0% relative relative lift over a 10.0% baseline requires 2,500 per arm at the standard a / power settings. Guardrails sit on revenue_per_user.

SAMPLE REQUIRED (PER ARM)

2,500

to detect a 2.0% relative relative effect at a = 0.05, power = 0.80

BASELINE
10.0%

DURATION
14 days

TOTAL SAMPLE
5,000

FEASIBILITY
VIABLE

MECHANISM

Why we think this works

Cart visitors have high intent. A complementary-product nudge at this moment converts more often than the same module on the PDP, based on the 2025Q3 recs experiment.

Cart sessions are only ~2k/day, which is the binding constraint on this test. AOV baseline ~$48. We knew this would be a tight powered experiment going in.

METRICS

conversion_rate is the verdict-driving metric.

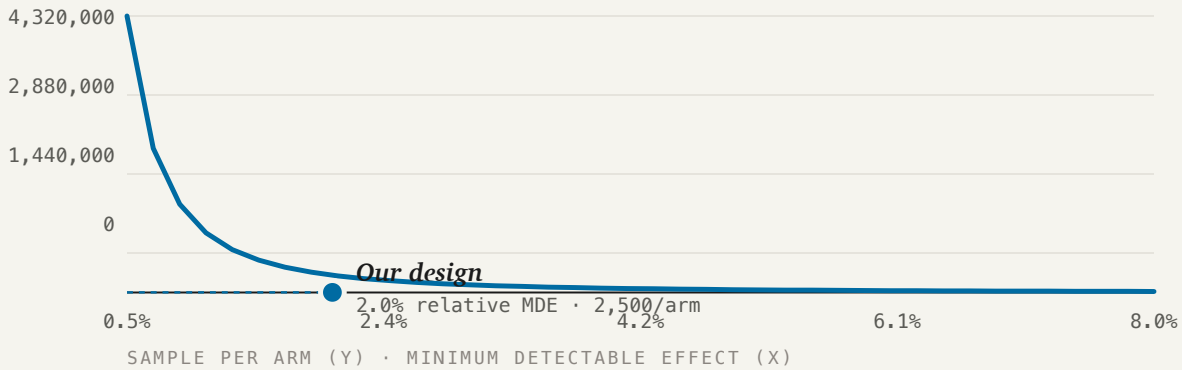
Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.

PRIMARY	**conversion_rate** Proportion · baseline 10.0% · MDE 2.0% relative · direction higher is better
SECONDARY	**conversion_rate** Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
GUARDRAIL	**revenue_per_user** Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 2.0% relative relative effect against a 10.0% baseline, we need 2,500 per arm. Tighter detection thresholds would climb the curve to the left — quickly.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the user_id level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.

UNIT	**user_id** One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed.
ARMS	**50 / 50** control · treatment
COHORT	**all assigned users**

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Ship if AOV CI excludes zero on benefit side. Likely to be underpowered — be prepared for an inconclusive outcome and a longer-running follow-up.

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width > 2 × MDE
INCONCLUSIVE	Underpowered (n < required) AND primary CI straddles zero
INVALID	SRM χ^2 p < 0.0005 and no override resolved — investigate assignment

INTEGRITY LOCK · THREE-PART SEAL

brief.sealed.yaml binds intent + metric definitions + expected shape.

Tampering with any of the three components after seal will fail `agentxp.schemas.brief_seal.verify_or_raise`; `agentxp analyze` refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.

DESIGN_CHAIN_HASH	bf842f93af13d54f6357c50cbb17d0431fb25010b8442e7f1d041f512c606857
METRIC_SNAPSHOT	1 metric hashed
CONVERSION_RATE	fbf7a228872bb052817eb550deb3311acba51c3cb60f03241c460211b307f2d0
EXPECTED_SHAPE	user_id · 50 / 50
SEALED_AT	2026-06-05T15:42:42Z
SEALED_BY	walk_experiment_demo